

COURSE PROJECT

Project Proposal

1 Title & Team

Project Title	<i>Community Needs Radar: Mapping Vulnerability and Service Accessibility in Greater Montreal</i>
Team Name	<i>Next Level</i>
Submission Date	<i>12 June 2026</i>

Team Member	Email	Program / Year
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2 Executive Summary

Community Needs Radar is an interactive Streamlit web app that maps social vulnerability against service access across Greater Montreal, surfacing neighborhoods where need is high, but services are sparse. The app offers two views from one shared dataset: a frontline worker view that generates service specific flyers for individuals, and an organization view that gives nonprofits, funders, and policymakers a map-based dashboard of vulnerability, service locations, and accessibility gaps

This matters because community need is spread unevenly across neighborhoods, while the data describing it sits fragmented across the 2021 Census, service directories like 211 Quebec, and municipal open data portals that non-technical users cannot easily combine. Without a simple way to compare need against access, frontline staff rely on memory and manual searches, and planners fall back on intuition or spreadsheets when deciding where to direct outreach, partnerships, and funding. The result is missed outreach, duplicated effort in already served areas, and under recognition of communities carrying the highest need. It also matters for the people seeking help directly: for those experiencing homelessness, who often lack reliable internet or digital access, a printed flyer of relevant nearby services makes support far easier to reach and to share. The tool serves community organizations, borough planners, nonprofit teams, funders, and policy analysts, as well as the vulnerable residents they aim to reach.

The expected outcome is a working web app delivering both paths end to end: a frontline view where workers generate ready to print flyers of relevant nearby services, and an organization view that combines public census, geographic, and service data into vulnerability indicators, service locations, accessibility gaps, ranked priority areas, and plain language neighborhood summaries. A transparent scoring approach, pairing a vulnerability score and a service access score into a resulting gap score, shows not only where attention is needed but why. Success will be measured by whether a non-technical user in either view can understand which areas need attention first, why they are flagged, and what community response may be appropriate, without merging a single dataset themselves.

3 Problem Statement

Community organizations and local planners often need to decide where to focus outreach, partnerships, funding, or service expansion. However, the information needed for these decisions is usually fragmented. Demographic data may be available from Statistics Canada, service information may be listed in separate

directories such as 211 Quebec, and geographic boundaries may come from municipal open data portals. These sources are valuable, but they are not always easy for non-technical users to combine into a single planning view.

The problem this project addresses is the lack of a simple, user-friendly tool that connects two important questions: where vulnerable communities are located, and whether those communities have accessible nearby services. A neighborhood may show high levels of social need based on income, age, language, immigration, household, or housing indicators. But without mapping service access alongside those indicators, it is difficult to identify which places may represent true planning gaps.

This problem affects community organizations, borough-level planners, nonprofit teams, funders, and policy analysts who need to make practical decisions with limited time and resources. The cost of leaving the problem unsolved is not only analytical. It can lead to missed outreach opportunities, uneven service planning, duplicated efforts in already-served areas, and under-recognition of communities that may need additional support.

The project will use publicly available data to build a dashboard that makes this issue visible and actionable. The goal is not to claim that distance alone fully defines access, nor to judge the quality of services. Instead, the project will provide a transparent and reproducible first layer of analysis: combining vulnerability indicators, service locations, and accessibility proxies into a clear map and priority ranking.

4 Objectives & Scope

Objectives	The primary objective of this project is to develop an interactive dashboard that helps identify underserved communities in Greater Montreal by combining demographic vulnerability indicators with community service accessibility measures. Specific objectives include: 1. Analyze selected public demographic and socioeconomic indicators from the 2021 Statistics Canada Census to develop a vulnerability profile for census tracts in Greater Montreal by June 21, 2026. 2. Collect, clean, and integrate geographic boundary data and community service location data to create a service accessibility dataset by July 5, 2026. 3. Develop a Vulnerability Index, Service Accessibility Score, and Gap Score to identify areas with high social need and limited access to support services by July 12, 2026. 4. Conduct geospatial analysis and neighborhood comparisons to highlight underserved areas and key drivers of vulnerability by July 19, 2026. 5. Design and implement an interactive dashboard featuring maps, rankings, area profiles, and visual summaries to support data-driven decision-making by July 30, 2026.
In Scope	This project focuses on communities within Greater Montreal and relies exclusively on publicly available datasets, including Statistics Canada Census data, geographic boundary files, and community service location data. The dashboard will provide vulnerability mapping, service accessibility analysis, neighborhood rankings, and gap identification through interactive visualizations and user-friendly summaries. The project is intended as a decision-support tool for community organizations, researchers, and policymakers seeking to better understand local service needs and disparities.
Out of Scope	The project will not use private or personally identifiable information, predict vulnerability at the individual level, evaluate the quality of specific service providers, or provide real-time service availability. Referral management, case management, and advanced routing functionality are outside the scope of this project.

5 Data Sources

Source	Type / Format	Access & Availability	Notes
<i>Statistics Canada 2021 Census Profile</i>	<i>CSV / web table / downloadable census data</i>	<i>Public</i>	<i>Main source for demographic and socioeconomic indicators such as income, age, immigration, language, housing, education, household type, and labour variables. Final geography level to be confirmed after data exploration.</i>
<i>Statistics Canada 2021 Census Boundary Files</i>	<i>Shapefile / GeoJSON after conversion</i>	<i>Public</i>	<i>Used to map census geographies such as census tracts or dissemination areas. Required for joining demographic data to geographic polygons.</i>
<i>Montreal Open Data: Administrative Boundaries</i>	<i>GeoJSON / SHP</i>	<i>Public</i>	<i>Boundary layer for Montreal boroughs, related cities, and agglomeration areas. Useful for map display and aggregation.</i>
<i>211 Quebec / 211 Grand Montreal Service Directory</i>	<i>Web directory / map / possible downloadable directory</i>	<i>Public, access format to be confirmed</i>	<i>Preferred source for community, public, and parapublic organization locations. Feasibility depends on whether data can be downloaded or structured efficiently.</i>
<i>Montreal Open Data: Libraries, community centers, public facilities, shelters, recreation facilities, or similar service proxies</i>	<i>CSV / GeoJSON / API depending on dataset</i>	<i>Public</i>	<i>Fallback source if 211 data cannot be downloaded in a structured format. These datasets can serve as proxy access points for community support infrastructure.</i>
<i>Public transit stops or walking-distance proxy data</i>	<i>CSV / GTFS / GeoJSON</i>	<i>Public, optional</i>	<i>Used only if time allows to improve accessibility measurement beyond straight-line distance.</i>

6 Methodology

The project will use a two-mode product and analytics methodology. V1: Public / Frontline Service Navigator will focus on helping residents, newcomers, volunteers, and frontline workers find relevant community services.

V2: Planner / Funder Intelligence will focus on helping community organizations, borough planners, policymakers, and funders understand vulnerability, service coverage, and potential gaps.

First, the team will define the study area, target users, and core user journeys for both modes. In V1, users should be able to search for services, filter by category, view locations on a map, and identify nearby or reachable options. In V2, users should be able to view vulnerability patterns, service coverage, priority areas, and explanation summaries.

Second, the team will collect and clean service-location data. The preferred source is a structured community service directory such as 211 Quebec. If structured access is limited, the team will use fallback public datasets such as libraries, community centers, shelters, clinics, or municipal service points. Service records will be standardized by name, category, address, coordinates, service type, and availability fields where possible.

Third, the team will create the accessibility layer. The MVP will use distance-based calculations or walking-distance proxies. If routing tools or free APIs are feasible, the dashboard will include 15/30/45-minute reachability zones. Otherwise, the team will document the distance proxy clearly as a limitation.

Fourth, the team will build the V2 planning analytics layer. Selected census indicators such as income, age, immigration, language, household, and housing variables will be normalized and combined into a transparent vulnerability score. The dashboard will show a heatmap, ranking table, and driver breakdown so users can understand why an area is flagged.

Fifth, the team will calculate service coverage and gap indicators by comparing vulnerability scores with service accessibility. Areas with higher vulnerability and lower reachable service coverage will be flagged as potential priority areas.

Finally, the team will validate the dashboard through data checks and usability testing. Validation will include coordinate checks, service-category review, formula review, map QA, and simple user tests with 2–3 non-technical users.

7 Deliverables

#	Deliverable	Description	Format
1	Final Report	<i>A comprehensive report covering project methodology, data sources, vulnerability index design, key findings, and recommendations, together with a user guide explaining how to navigate the dashboard, interpret scores, and use each feature. Intended for course instructors, community organizations, and policymakers.</i>	PDF
2	Web Application	<i>An interactive web application built with Streamlit, offering two user experiences: a Planner View for social workers, nonprofits, and policymakers (e.g., layered map with filters, vulnerability scoring, gap analysis, exportable reports) and a Community View for individuals in need (e.g., simplified service map, contact information). Both modes include an AI chatbot for plain-language queries and are available in English and French. Intended for community organizations, planners, funders, and residents of Greater Montreal.</i>	Web app (Streamlit)
3	GitHub Repository	<i>A public repository containing all source code, data processing notebooks, and pipeline scripts, organized with a clear folder structure and reproducibility instructions. Intended for developers, researchers, or future teams who want to reproduce, audit, or extend the project.</i>	GitHub
4	Technical Documentation	<i>A README and supporting data sources, vulnerability index construction, environment setup, feature descriptions, known limitations, and next-step recommendations. Intended for technical users and future contributors who need to understand, maintain, or build upon the system.</i>	Markdown (in repo)

#	Deliverable	Description	Format
5	Final Presentation	<i>A live presentation demonstrating the web, key findings, and design decisions, accompanied by a slide deck summarizing the project from problem to outcome.</i> <i>Intended for course instructors and classmates; optionally delivered to community stakeholders or partner organizations.</i>	Slides (PPT) + demo

8 Timeline / Gantt

Phase / Task	W1	W2	W3	W4	W5	W6	W7	W8
Phase 1 – Define & Plan								
Scope, roles & proposal writing								
Tech stack & repo setup								
✓ Milestone 1 — Proposal submitted, scope locked <i>Depends on: roles assigned, features finalized</i>								
Phase 2 – Data Collection & Analysis								
Census data collection & cleaning								
Boundary files, spatial join & service data								
Vulnerability index & gap score								
Temporal trends & cost planning model								
✓ Milestone 2 — All data cleaned & joined, vulnerability index defined <i>Depends on: raw data downloaded and cleaned, spatial join verified</i>								
Phase 3 – Web App Development & Testing								
Dashboard / web app structure & layout								
Service locator & map UI								
AI chatbot & neighborhood summary								
User modes, alerts & EN/FR toggle								
✓ Milestone 3 — MVP complete, both modes functional <i>Depends on: all data ready, vulnerability index built, dashboard structure set up</i>								
Validation, user testing & bug fixes								
<i>Buffer — W7 includes buffer time for final revisions and unexpected delays</i>								
Phase 4 – Write-up & Deliverables								
Final report & technical documentation								
Presentation prep & final demo								

Phase / Task	W1	W2	W3	W4	W5	W6	W7	W8
✓ Milestone 4 — All deliverables submitted, final presentation delivered <i>Depends on: validation complete, bug fixes applied</i>								
Reports								
Project Proposal (Jun 12)								
Progress Report 1 (Jun 21)								
Progress Report 2 (Jul 5)								
Progress Report 3 (Jul 19)								
Progress Report 4 (Jul 26)								
Final Presentation (Jul 30)								

9 Risks

Risk	Likelihood	Impact	Mitigation
Data availability & completeness — key datasets may be incomplete or inaccessible	<i>M</i>	<i>H</i>	Upon initial review of key datasets, some sources may have limited accessibility or incomplete structure. Fallback sources have been identified, including Montreal Open Data public facilities (e.g., libraries, shelters, community centers) as proxy service points. Adjust scope early to avoid downstream delays.
Subjectivity in index definition — no universally accepted standard for "high-need" or vulnerability levels; indicators and cutoffs must be self-defined, which may be questioned	<i>M</i>	<i>M</i>	Document all methodological decisions transparently in the report, including the rationale and references used to define each indicator and threshold. Acknowledge limitations explicitly.
AI accuracy & hallucination — open-source LLM may produce inaccurate summaries or chatbot responses due to limited model capability	<i>M</i>	<i>M</i>	Ground all outputs strictly in structured dashboard data via RAG. Restrict chatbot to factual lookups only. Evaluate model options early in W4. Add a disclaimer in the UI that AI responses are for reference only.
Spatial data join failure — census demographic data and boundary files may use inconsistent geographic identifiers, making it difficult for mapping.	<i>M</i>	<i>H</i>	Standardize geographic identifiers across all datasets from the start. Test data join in W4 before building other features. If needed, fall back to borough-level data instead of census tract level.

10 Roles

Team Member	Primary Responsibilities	Backup For
Chloe	Product / project lead: scope, timeline, proposal coordination, professor communication, MVP decisions	AI/ RAG Part + Dashboard/app
Laura	Data engineering lead: download, clean, merge, and document census, boundary, and service datasets	AI/ RAG + Geospatial + Dashboard/App

Team Member	Primary Responsibilities	Backup For
Frondy	Geospatial / analytics lead: map joins, vulnerability score, service access score, gap score	Github and app documentation
Jessie	Dashboard / app lead: design and build the Streamlit web dashboard, implement two-mode UI, manage overall UX flow	AI/ RAG + Geospatial + Github
Mariam	AI / insight / presentation lead: area summaries, methodology wording, user testing, final presentation story	

11 Success Metrics

Metric	Definition	Target	How Measured
<i>Service navigation usability</i>	<i>Public/frontline users can find relevant services through the dashboard</i>	<i>User can find one relevant nearby service in under 2 minutes</i>	<i>Usability test</i>
<i>Accessibility layer</i>	<i>Platform provides a distance or travel-time proxy for reachable services</i>	<i>Distance proxy or 15/30/45-minute zones implemented</i>	<i>Dashboard QA</i>
<i>Vulnerability analysis</i>	<i>Platform shows high-need areas with score, ranking, and driver breakdown</i>	<i>At least 3–5 census indicators included and documented</i>	<i>Data dictionary + dashboard</i>
<i>Gap identification</i>	<i>Platform identifies areas with high vulnerability and lower service coverage</i>	<i>Top 10 potential priority areas generated</i>	<i>Gap score output</i>
<i>Technical reproducibility</i>	<i>Data processing and dashboard can be reproduced</i>	<i>GitHub repo includes README, methodology, and setup instructions</i>	<i>Repo review</i>
<i>Community usefulness</i>	<i>Non-technical users can understand and use the dashboard</i>	<i>At least 2–3 external users can complete a simple task without major guidance</i>	<i>Short usability test</i>